

JEE MAIN REVISION MASTER: VECTOR ALGEBRA

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 282 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Dot Product, Projections & Orthogonality	191 PYQs	Scalar Product
Very High	Cross Product & Geometric Areas	184 PYQs	Vector Product
Very High	Scalar Triple Product (Box Product) & Coplanarity	55 PYQs	Volume & Coplanarity
Very High	Section Formula, Collinearity & Centroids	38 PYQs	Geometry & Position Vectors
Very High	Vector Triple Product & Lagrange Identity	31 PYQs	Triple Products
Very High	Linear Combination & Vector Equations	30 PYQs	Vector Equations
High	Vector Algebra - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Dot Product, Projections & Orthogonality	$\vec{a} \cdot \vec{b} = \vec{a} \vec{b} \cos\theta$, $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$, $\text{Proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{ \vec{b} ^2} \vec{b}$
Cross Product & Geometric Areas	$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$, $\text{Area}_{\triangle} = \frac{1}{2} \vec{a} \times \vec{b} $, $\hat{n} = \frac{\vec{a} \times \vec{b}}{ \vec{a} \times \vec{b} }$
Scalar Triple Product (Box Product) & Coplanarity	$[\vec{a}; \vec{b}; \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}$, $[\vec{a}; \vec{b}; \vec{c}] = 0 \iff \text{coplanar}$
Section Formula, Collinearity & Centroids	$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$, $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$, $\vec{a} = \lambda \vec{b} \iff \text{collinear}$
Vector Triple Product & Lagrange Identity	$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$, $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$
Linear Combination & Vector Equations	$\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$, $\text{or take } \vec{a} \times \vec{r} \text{ to decouple}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Dot Product, Projections & Orthogonality (Very High Yield — 191 PYQs)

Core Principle: Dot product $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos(\theta) = a_x b_x + a_y b_y + a_z b_z$. Condition for perpendicular vectors: $\mathbf{a} \cdot \mathbf{b} = 0$. Scalar projection of $\mathbf{a}$ on $\mathbf{b}$ is $(\mathbf{a} \cdot \mathbf{b}) / |\mathbf{b}|$.

■■■ **Caution / JEE Pitfall:** 1. Projection vector along $\mathbf{b}$ is $((\mathbf{a} \cdot \mathbf{b}) / |\mathbf{b}|^2) * \mathbf{b}$. 2. Modulus square $|\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 + 2(\mathbf{a} \cdot \mathbf{b})$. Do not write $2|\mathbf{a}||\mathbf{b}|$ unless vectors are collinear!

■ **Speed Trick:** To find unit vector bisecting the angle between $\mathbf{a}$ and $\mathbf{b}$: $(\hat{\mathbf{a}} + \hat{\mathbf{b}}) / |\hat{\mathbf{a}} + \hat{\mathbf{b}}|$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 10\hat{i} + 2\hat{j} - \hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b}) + 3(\dots)$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let $\mathbf{u}$ and $\mathbf{v}$ be unit vectors inclined at an acute angle such that $|\mathbf{u} \times \mathbf{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \dots$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let P be a point in the plane of the vectors $\vec{AB} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{AC} = \hat{i} - \hat{j} + 3\hat{k}$ such that P is equi..." [Ans: N/A]

3.2 Cross Product & Geometric Areas (Very High Yield — 184 PYQs)

Core Principle: Cross product $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}|\sin(\theta)\hat{n}$. Condition for collinear/parallel vectors: $\mathbf{a} \times \mathbf{b} = 0$. Area of parallelogram with sides $\mathbf{a}$, $\mathbf{b}$ is $|\mathbf{a} \times \mathbf{b}|$; with diagonals $\mathbf{d}_1$, $\mathbf{d}_2$ is $(1/2)|\mathbf{d}_1 \times \mathbf{d}_2|$.

■■■ **Caution / JEE Pitfall:** 1. Cross product is anti-commutative: $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$. 2. Lagrange's identity: $|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2$.

■ **Speed Trick:** Area of parallelogram given diagonals $\mathbf{d}_1$ and $\mathbf{d}_2$ is $(1/2)|\mathbf{d}_1 \times \mathbf{d}_2|$, NOT $|\mathbf{d}_1 \times \mathbf{d}_2|$!

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 10\hat{i} + 2\hat{j} - \hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b}) + 3(\dots)$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let $\mathbf{u}$ and $\mathbf{v}$ be unit vectors inclined at an acute angle such that $|\mathbf{u} \times \mathbf{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \dots$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let P be a point in the plane of the vectors $\vec{AB} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{AC} = \hat{i} - \hat{j} + 3\hat{k}$ such that P is equi..." [Ans: N/A]

3.3 Scalar Triple Product (Box Product) & Coplanarity (Very High Yield — 55 PYQs)

Core Principle: Box product $[a \ b \ c] = a \cdot (b \times c)$ = determinant of components. Three vectors are coplanar if and only if $[a \ b \ c] = 0$. Volume of parallelepiped = $|[a \ b \ c]|$; volume of tetrahedron = $(1/6) \cdot |[a \ b \ c]|$.

■ **Caution / JEE Pitfall:** 1. Cyclic permutation maintains value: $[a \ b \ c] = [b \ c \ a] = [c \ a \ b]$. Non-cyclic permutation negates value: $[a \ c \ b] = -[a \ b \ c]$. 2. If any two vectors are equal or parallel, $[a \ b \ c] = 0$.

■ **Speed Trick:** Four points A, B, C, D are coplanar if $[AB \ AC \ AD] = 0$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = i\hat{j} + 2k\hat{j}$, $\vec{b} = i\hat{j} - 3k\hat{j}$, $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d} = \vec{c} \times \vec{a}$..." [Ans: N/A]
- [2025 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = i\hat{j} + 2j\hat{j} + 3k\hat{j}$, $\vec{b} = 3i\hat{j} + j\hat{j} - k\hat{j}$ and $\vec{c}$ be three vectors such that $\vec{c}$ is coplanar with..." [Ans: N/A]
- [2025 Morning Shift] Q#3 (MCQ): "Let the position vectors of the vertices A, B and C of a tetrahedron ABCD be $i\hat{j} + 2j\hat{j} + k\hat{j}$, $i\hat{j} + 3j\hat{j} - 2k\hat{j}$ and $2i\hat{j}$..." [Ans: N/A]

3.4 Section Formula, Collinearity & Centroids (Very High Yield — 38 PYQs)

Core Principle: Section formula $r = (m \cdot b + n \cdot a) / (m + n)$. Three points A, B, C are collinear if $AB = \lambda \cdot BC$ or $AB \times AC = 0$. Centroid of triangle $G = (a + b + c) / 3$.

■ **Caution / JEE Pitfall:** Collinear vectors can have opposite directions (negative lambda). External division has a minus sign: $(mb - na) / (m - n)$.

■ **Speed Trick:** In a triangle, line joining midpoints is parallel to third side and half its length: $MN = (1/2) \cdot BC$.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "Let the lines $L_1: \vec{r} = i\hat{j} + 2j\hat{j} + 3k\hat{j} + \lambda(2i\hat{j} + 3j\hat{j} + 4k\hat{j})$, $\lambda \in \mathbb{R}$ and $L_2: \vec{r} = 4i\hat{j} + \hat{j} + \hat{t} \dots$ " [Ans: N/A]
- [2026 Evening Shift] Q#3 (MCQ): "Let O be the origin, $\vec{OP} = \vec{a}$ and $\vec{OQ} = \vec{b}$. If R is the point on $\vec{OP}$ such that $\vec{OP} = 5\vec{OR}$, and $\dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#3 (Numerical): "Let PQR be a triangle such that $\vec{PQ} = -2i\hat{j} - j\hat{j} + 2k\hat{j}$ and $\vec{PR} = ai\hat{j} + bj\hat{j} - 4k\hat{j}$, $a, b \in \mathbb{Z}$. Let S be the p..." [Ans: 37]

3.5 Vector Triple Product & Lagrange Identity (Very High Yield — 31 PYQs)

Core Principle: Vector triple product expands via the BAC-CAB rule: $a \times (b \times c) = (a \cdot c) \cdot b - (a \cdot b) \cdot c$. It lies in the plane of b and c, perpendicular to a.

■ **Caution / JEE Pitfall:** Vector triple product is NOT associative: $a \times (b \times c) \neq (a \times b) \times c$. Pay rigorous attention to bracket positioning!

■ **Speed Trick:** Mnemonic: 'BAC minus CAB'. Dot product of outer vectors multiplied by inner vector, minus dot product of first two multiplied by third.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = i\hat{j} + 2j\hat{j} + 3k\hat{j}$, $\vec{b} = i\hat{j} - 3k\hat{j}$, $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d} = \vec{c} \times \vec{a}$..." [Ans: N/A]
- [2025 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = i\hat{j} + 2j\hat{j} + 3k\hat{j}$, $\vec{b} = 3i\hat{j} + j\hat{j} - k\hat{j}$ and $\vec{c}$ be three vectors such that $\vec{c}$ is coplanar with..." [Ans: N/A]
- [2024 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = -5i\hat{j} + j\hat{j} - 3k\hat{j}$, $\vec{b} = i\hat{j} + 2j\hat{j} - 4k\hat{j}$ and $\vec{c} = ((\vec{a} \times \vec{b}) \times \vec{c}) \times \vec{c}$..." [Ans: N/A]

3.6 Linear Combination & Vector Equations (Very High Yield — 30 PYQs)

Core Principle: Solving vector equations of the form $\mathbf{r} \times \mathbf{a} = \mathbf{b}$ and $\mathbf{r} \cdot \mathbf{c} = d$. Express $\mathbf{r}$ as a linear combination or take cross/dot products on both sides with known vectors.

■ **Caution / JEE Pitfall:** When $\mathbf{r} \times \mathbf{a} = \mathbf{b}$, note that $\mathbf{b}$ must be perpendicular to $\mathbf{a}$ ($\mathbf{b} \cdot \mathbf{a} = 0$). Then $\mathbf{r} \times \mathbf{a} = \mathbf{b}$ implies $\mathbf{r} = \frac{\mathbf{a} \times \mathbf{b}}{|\mathbf{a}|^2} + \lambda \mathbf{a}$.

■ **Speed Trick:** If $\mathbf{r} \times \mathbf{a} = \mathbf{b}$, taking dot product with $\mathbf{c}$ gives $[\mathbf{r} \ \mathbf{a} \ \mathbf{c}] = \mathbf{b} \cdot \mathbf{c}$. Taking cross product with $\mathbf{a}$ helps resolve $\mathbf{r}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 10\hat{i} + 2\hat{j} - \hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b}) + 3(\dots)$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let $\mathbf{u}$ and $\mathbf{v}$ be unit vectors inclined at an acute angle such that $|\mathbf{u} \times \mathbf{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \dots$ " [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Let the vectors $\vec{a} = -\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$. For some $\lambda, \mu \in \mathbb{R}$, let $\vec{c} = \lambda \mathbf{a} + \mu \mathbf{b}$ " [Ans: N/A]

3.7 Vector Algebra - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Vector Algebra.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (Numerical): "If the distance of the point $P(43, \alpha, \beta)$, $\beta < 0$, from the line $\vec{r} = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k})$, $\mu \in \mathbb{R}$ is $\dots$ " [Ans: 170]
- [2025 Morning Shift] Q#3 (MCQ): "Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + 7\hat{j} + 3\hat{k}$. Let $\vec{r} = -\hat{i} + 2\hat{j} + \hat{k} + \lambda \vec{a}$ " [Ans: N/A]
- [2021 Evening Shift] Q#4 (Numerical): "For $p > 0$, a vector $\vec{v}_2 = 2\hat{i} + (p+1)\hat{j}$ is obtained by rotating the vector $\vec{v}_1 = \sqrt{3}p\hat{i} + \hat{j}$ by an angle $\dots$ " [Ans: 6]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{j} - \hat{k}$ and $\vec{c}$ be three vectors such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{c} \cdot (\vec{a} - 2\vec{b})$ is equal to ____.

Correct Answer: 3

Solution Analysis: Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{j} - \hat{k}$. We have to find $\vec{c} \cdot (\vec{a} - 2\vec{b})$ given that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$. Let $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$. Then $\vec{a} = (1, 1, 1)$, $\vec{b} = (0, 1, -1)$. Now compute $\vec{a} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ x & y & z \end{vmatrix}$. Expanding, $\vec{a} \times \vec{c} = \hat{i}(z-y) - \hat{j}(z-x) + \hat{k}(y-x)$. Since this is equal to $\vec{b} = \hat{j} - \hat{k}$, we compare components: $z-y=0$, $-(z-x)=1$, $y-x=-1$. $\Rightarrow$

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Evening Shift (Q#2)

If the distance of the point $P(43, \alpha, \beta)$, $\beta < 0$, from the line $\vec{r} = 4\hat{i} - k\hat{j} + \mu(2\hat{i} + 3k\hat{j})$, $\mu \in \mathbb{R}$ along a line with direction ratios $3, -1, 0$ is $13\sqrt{10}$, then $\alpha^2 + \beta^2$ is equal to _____

Correct Answer: 170

Solution Analysis: Point $P : (43, \alpha, \beta)$ with the constraint $\beta < 0$. Target Line (L_1) : Given by $\vec{r} = 4\hat{i} - k\hat{j} + \mu(2\hat{i} + 3k\hat{j})$. A general point Q on this line is $(4 + 2\mu, 0, -1 + 3\mu)$. Direction of measurement : The distance is measured along a line with direction ratios $(3, -1, 0)$. Find Point Q on the Target Line Since the distance is measured along the direction $d = (3, -1, 0)$, the vector PQ must be parallel to d . $PQ = Q - P = (4 + 2\mu - 43, 0 - \alpha, -1 + 3\mu - \beta)$ $PQ = (2\mu - 39, -\alpha, 3\mu - \beta - 1)$ $PQ = (2\mu - 39, -\alpha, 3\mu - \beta - 1)$

Benchmark Problem 3: JEE Main 2026 (Online) 28th January Morning Shift (Q#3)

Let PQR be a triangle such that $\vec{PQ} = -2\hat{i} - j\hat{j} + 2k\hat{k}$ and $\vec{PR} = a\hat{i} + bj\hat{j} - 4k\hat{k}$, $a, b \in \mathbb{Z}$. Let S be the point on QR , which is equidistant from the lines PQ and PR . If $|\vec{PR}| = 9$ and $\vec{PS} = \hat{i} - 7j\hat{j} + 2k\hat{k}$, then the value of $3a - 4b$ is _____

Correct Answer: 37

Solution Analysis: We are given that $|\vec{PR}| = 9$ Using the distance formula for the vector $\vec{PR} = a\hat{i} + bj\hat{j} - 4k\hat{k}$: $\sqrt{(a^2) + b^2 + (-4)^2} = 9 \Rightarrow a^2 + b^2 + 16 = 81 \Rightarrow a^2 + b^2 = 65$ The problem states that point S lies on QR and is equidistant from lines PQ and PR . In a triangle, the locus of points equidistant from two sides is the angle bisector. Therefore, $\vec{PS}$ is the internal angle bisector of $\angle QPR$. For an angle bisector $\vec{PS}$, it must be collinear with the sum of the unit vectors along the adjacent sides : $\vec{PS} = \lambda \left(\frac{\vec{PQ}}{|\vec{PQ}|} + \frac{\vec{PR}}{|\vec{PR}|} \right)$